

# Object Detection and Localization Sight to Sound Framework for Visually Impaired Community

**Abstract**—This study tackles a crucial yet often overlooked challenge in the daily lives of visually impaired individuals, shifting the focus beyond traditional mobility aids. With 253 million people worldwide affected by visual impairment, tasks such as object localization remain significant obstacles to independence. This research presents a framework designed specifically to assist visually impaired individuals in locating objects based on sound. It introduces two innovative methods: the Rotated Clock Directioning System and the Hand-Relative Directioning System, both leveraging advanced machine learning techniques. The framework integrates an optimized YOLOv8 model for precise object detection and localization, translating the results into speech using advanced speech recognition methods. The system was tested for object detection both from the frame center and in relation to the hand's position. Testing was conducted on a diverse set of household objects, and the framework showed exceptional effectiveness with competitive results. This highlights its potential to improve object localization, thereby significantly enhancing the autonomy and quality of life for visually impaired individuals.

**Keywords**—Visually impaired people, healthcare, assistive devices, sensor directioning systems, object detection and localization

## I. INTRODUCTION

Globally, over 253 million people live with visual impairments, including 36 million who are completely blind [1]. According to the World Health Organization, 90% of the visually impaired (VI) population resides in low-income regions of developing countries [2]. In response to this widespread challenge, assistive technology was introduced in the 1960s, primarily aimed at addressing orientation and navigation difficulties for the visually impaired [3]. The capabilities of assistive technology can generally be categorized into three key areas: communication with the environment, interaction with devices, and engagement with other people [4].

Despite the development and commercialization of various assistive technologies, a crucial gap remains—none of the available solutions effectively assist VI individuals in locating specific household items or identifying objects in their surroundings. For example, [5] proposed an obstacle avoidance system using a belt device equipped with a Kinect depth sensor to detect and avoid obstacles by constructing a 3D model of the environment. Similarly, [6] introduced a smart cane with notifications that provide users with a more comprehensive understanding of their surroundings compared to a traditional cane, enhancing the safety and independence of VI users.

However, these technologies focus on obstacle avoidance and mobility, leaving the vital issue of object localization unaddressed. In this paper, we propose a groundbreaking solution to this problem: Sight to Sound Framework the first self-contained system designed to help VI individuals search for and find objects within their environment. Our framework tackles the neglected need for object localization through the

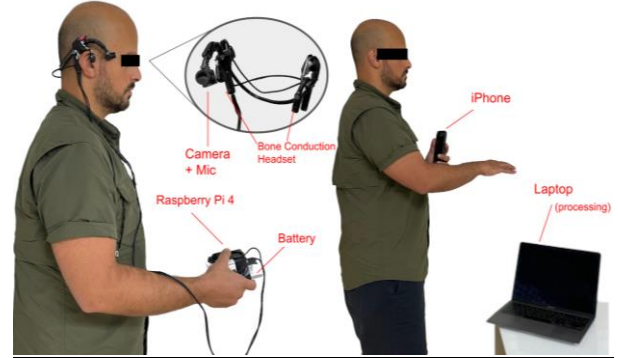


Fig. 1. The left side shows the user wearing the head mounted Sight to Sound System, while the right side demonstrates its use with an iPhone and laptop.

introduction of two novel methods. The first is the Rotated Clock Directioning System, which provides user-friendly spatial instructions, guiding users through a 3D space. The second is the Hand-Relative Directioning System, which allows users to accurately move their hands toward the desired object.

These directioning methods are powered by state-of-the-art machine learning models, including an optimized "You Only Look Once" YOLOv8 [7], [8] variant for object detection, specifically tailored for assisting visually impaired individuals in everyday scenarios. The YOLOv8 model was trained to recognize and localize common household objects, enabling users to navigate their environment more independently. The training process involved dataset of images representing various household environments, such as kitchens, living rooms, and bedrooms. Objects used for training was frequently used items like keys, remote controls, glasses, and phones, as well as larger items like furniture and appliances.

To refine the model for real-world use, additional datasets were added to simulate scenarios like room entry/exit, doorway detection, and obstacle identification, enhancing spatial awareness. YOLOv8 was optimized for fast, low-cost detection on portable devices like Raspberry Pi and wearable cameras, essential for visually impaired users. Integrated with VOSK-API [9] for real-time speech recognition, users can issue voice commands (e.g., "Where are my keys?"). The model then scans the environment, identifies targets, and provides audio guidance via the Rotated Clock or Hand-Relative Directioning System. The rest of the paper is organized to present the "Sight to Sound Framework," detailing its components, object localization and spatial guidance methods, experimental results, and performance analysis.

## II. MATERIALS AND METHODS

Our assistive framework integrates machine learning and sensor technologies into a user-friendly system. It performs two main tasks: (1) object detection and localization using a

modified YOLOv8 model (Section II.A) and (2) user guidance through innovative directioning methods (Section II.B). The framework runs in real time on portable devices like laptop-smartphones and Raspberry Pi. It includes two key modules: the Rotated Clock Directioning System for 3D spatial guidance and the Hand-Relative Directioning System for precise hand movement. It processes about 10 frames per second, assisting visually impaired users in daily tasks. Figures 1 and 2 illustrate the hardware setup and software pipeline.

#### A. Object Detection

The proposed framework is built upon a variant of the YOLOv8 object detection model, optimized for efficient and real-time object localization from a camera feed. The pipeline processes continuous video sequence  $V$ , comprising frames  $I_t$  at time  $t$ , through stages of preprocessing, feature extraction, multi-scale feature aggregation, and object detection, yielding bounding box coordinates and class predictions in real time. Each frame  $I_t$  from the camera feed is adaptively scaled to align with the model's input requirements:

$$I'_t = S(I_t, s) \quad (1)$$

where  $I'_t$  is the scaled frame,  $S(\cdot)$  denotes the scaling function, and  $s$  is the scaling factor (e.g., resizing to  $640 \times 640$  pixels). During training, mosaic data augmentation enhances generalization by combining multiple frames:

$$I_{aug} = M(I_{1t}', I_{2t}', I_{3t}', I_{4t}') \quad (2)$$

where  $M(\cdot)$  merges four scaled frames  $I_{1t}', I_{2t}', I_{3t}', I_{4t}'$  into a single augmented frame  $I_{aug}$ , though this is applied only during training, not inference from the live feed. The backbone network, typically a CSPDarknet variant, extracts hierarchical feature maps from each processed frame:

$$F_t = \mathcal{B}(I'_t) \quad (3)$$

where  $\mathcal{B}(\cdot)$  represents the backbone's convolutional layers, producing feature maps  $F_t$  at multiple resolutions (e.g.,  $80 \times 80$ ,  $40 \times 40$ ,  $20 \times 20$ ) to capture spatial and contextual details in real time. YOLOv8's anchor-free design ensures direct feature extraction without additional tensor-slicing modules. A neck module, integrating a Feature Pyramid Network (FPN) and Path Aggregation Network (PAN), fuses multi-scale features to enable detection across object sizes:

$$F_{neck,t} = \mathcal{FPN}(F_t) + \mathcal{PAN}(F_t) \quad (4)$$

where:  $\mathcal{FPN}(F_t)$  enhances top-down semantic propagation, and  $\mathcal{PAN}(F_t)$  strengthens bottom-up detail flow, resulting in a robust feature representation  $F_{neck,t}$  for each frame. The detection head processes  $F_{neck,t}$  to predict bounding boxes, objectness scores, and class probabilities in a single forward pass:

$$D_t = \mathcal{H}(F_{neck,t}) \quad (5)$$

where  $\mathcal{H}(\cdot)$  is the detection function, and  $D_t = \{(b_{i,t}, c_{i,t}, p_{i,t})_{i=1}^N\}$  denotes the detected objects at time  $t$ . Here:  $b_{i,t} = x_{i,t}, y_{i,t}, w_{i,t}, h_{i,t}$  represents the bounding box coordinates (center  $x_{i,t}, y_{i,t}$ , width  $w_{i,t}$ , height  $h_{i,t}$ ),  $c_{i,t}$  is the class label, and  $p_{i,t}$  is the confidence score. These coordinates are computed as:

$$x_{i,t} = (\text{sigmoid}(t_{x_{i,t}}) + c_{x_{i,t}}) \cdot \text{stride} \quad (6)$$

$$y_{i,t} = (\text{sigmoid}(t_{y_{i,t}}) + c_{y_{i,t}}) \cdot \text{stride} \quad (7)$$

$$w_{i,t} = p_w \cdot e^{t_{w_{i,t}}} \quad (8)$$

$$h_{i,t} = p_h \cdot e^{t_{h_{i,t}}} \quad (9)$$

where  $t_{x_{i,t}}, t_{y_{i,t}}, t_{w_{i,t}}, t_{h_{i,t}}$  are raw network outputs,  $c_{x_{i,t}}, c_{y_{i,t}}$  are grid offsets, and  $\text{stride}$  adjusts for the feature map's scale.

The YOLOv8 model is optimized during training to minimize a composite loss function, ensuring accuracy and efficiency for real-time camera feed processing:

$$\mathcal{O}_{opt} = \min_{\theta} [\sum_{i=1}^N \mathcal{L}(b_{i,t}, c_{i,t}, p_{i,t}) + \lambda \|\theta\|^2] \quad (10)$$

where:  $\mathcal{L}$  is the total loss function (including classification, localization, and object confidence losses),  $\lambda \|\theta\|^2$  is the regularization term to prevent overfitting.

#### B. Specifications Speech Recognition system

To implement the Sight to Sound Framework, we employ the VOSK-API [9], an open-source speech recognition toolkit designed for offline use, making it particularly useful for applications with limited or no internet access. VOSK supports a wide array of languages and is compatible with multiple platforms, including Windows, Linux, and macOS. Its core strength lies in its ability to deliver real-time speech recognition with high accuracy while requiring minimal

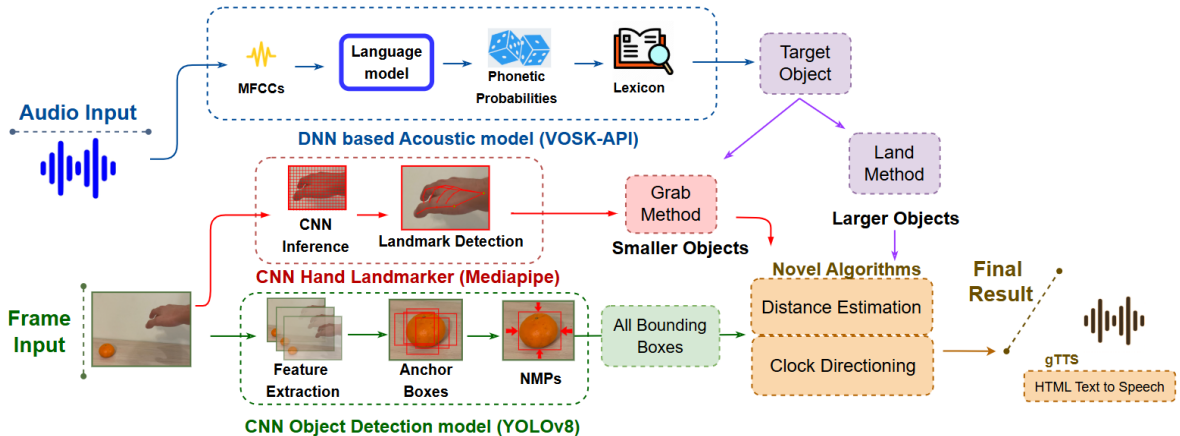


Fig. 2. Sight to Sound Framework pipeline is depicted, beginning with audio and frame input at the top and ending with the final audio output at the bottom left. The detailed walkthrough of the system pipeline as depicted in this figure is in figure 2, 3 and 4.

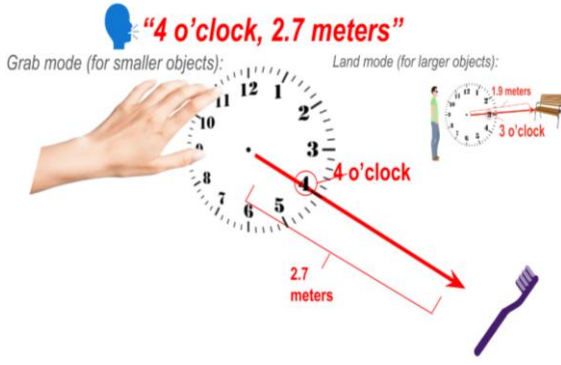


Fig. 3. Directioning system: The left side shows grab mode, while the top right displays land mode. Audio feedback includes clock degrees and forward distance in meters.

computational resources, making it ideal for resource-constrained environments and enhancing the portability of the Sight to Sound Framework. Once an object is detected and localized, the VOSK-API converts speech commands into text. The speech signal  $S(t)$  is processed as Compute Mel-Frequency Cepstral Coefficients (MFCCs):

$$M_k = \sum_{m=1}^M \log(E_m) \cos[k(m - 0.5) \frac{\pi}{M}] \quad (11)$$

where  $E_m$  is the Mel-scale energy. We use a Hidden Markov Model (HMM) to determine the most probable words  $W$ :

$$P(W | M) = \arg \max P(W)P(M | W) \quad (12)$$

The recognized text is matched with detected objects from YOLO's bounding boxes. For example: If the user says "Where is the chair?", the system retrieves the bounding box of the chair and provides an audio response like: "The chair is at 2 o'clock, 1.5 meters away." This capability ensures effective performance in the Sight to Sound Framework, making it a valuable tool for the visually impaired community by providing seamless voice interaction and support in navigating their environments, ultimately enhancing their independence.

### C. Rotated Clock Directioning System

To ensure that the Sight to Sound Framework is accessible and easy to use for visually impaired individuals, we implement the Rotated Clock Directioning System. This innovative approach provides spatial guidance and orientation by utilizing a clock face analogy to convey directional information in a way that is both intuitive and user-friendly. In this framework, directions are represented on a clock face, with "12 o'clock" indicating straight ahead and "3 o'clock" pointing to the right. Directions are represented as angles relative to the user's current orientation:

$$\theta = \frac{360^\circ}{12} \times h \quad (13)$$

Where  $\theta$  is the directional angle,  $h$  is the clock position (1 to 12 o'clock). For real-time adaptation, the system dynamically updates relative to the user's facing direction  $\phi$ , adjusting the final directional cue:

$$\theta' = (\theta + \phi) \bmod 360^\circ \quad (14)$$

where:  $\phi$  is the user's current heading direction. This ensures that directional indicators always remain contextually relevant. The system integrates with YOLO-based object detection and speech recognition, allowing detected objects to be localized using:

$$D = r \cdot e^{i\theta'} \quad (15)$$

where:  $D$  is the object's relative position,  $r$  is the distance to the object,  $\theta'$  is the adjusted directional angle. Additionally, it integrates seamlessly with other assistive technologies, such as object detection and speech recognition, allowing for precise location guidance. This combination makes the framework an effective tool for improving spatial orientation and fostering greater independence among visually impaired users.

### D. Inferring Distance by Relativity

The Sight to Sound Framework delivers audio directions in a 3D space, unlike most assistive devices that work on a 2D plane, like navigation apps. This 3D approach matters because objects can be above or below, not just forward, backward, left, or right, enhancing usability and accuracy for visually impaired (VI) users. We've added a unique feature: directional guidance relative to the user's hand, which helps with small objects like a fork, bottle, or toothbrush that a VI user might want to pick up. Larger objects, such as a bench, bus, or car, don't need this hand-relative guidance. This is a groundbreaking advancement in VI assistive tech, the first of its kind (see figure 3). We also included an environment feature, letting users ask about all objects around them, tested thoroughly for reliability.

For object detection, we use OpenCV to get coordinates of two opposite corners of a rectangular bounding box around each object. By finding the difference between these coordinates, we calculate the object's length and width. To measure the live distance from the camera, we compare the object's current length and width to stored ratios (based on measurements at a known 1-meter distance). These ratios are kept in a dictionary for reference. After detection, we take the live measurements from OpenCV, compare them to the stored ratios, and calculate two distance estimates—one from length, one from width. To ensure accuracy despite possible rotation skewing, we average these two distances, providing precise spatial information to the user (see figure 4).

$$Distance = \frac{\frac{|x_1 - x_2|}{m}}{|x'_1 - x'_2|} + \frac{\frac{|y_1 - y_2|}{m}}{|y'_1 - y'_2|} / 2 \quad (16)$$

Where  $(x_1, y_1), (x_2, y_2)$  are the prerecorded corner coordinates,  $m$  denote prerecorded distance, and  $(x'_1, y'_1), (x'_2, y'_2)$  denote the live received corner coordinates see figure 5.

### E. Hand Recognition and Proximity Detection

To determine whether a user's hand is near an object, we employ MediaPipe Hands, which detects key landmarks on the hand. Given the detected object's bounding box  $b = (x_1, y_1, x_2, y_2)$  the framework calculates the Euclidean distance between the object's center and the palm center:

$$d_{hand,obj} = \sqrt{(x_{hand} - x_{obj})^2 + (y_{hand} - y_{obj})^2} \quad (17)$$

Where  $(x_{hand}, y_{hand})$  represents the palm center and  $(x_{obj}, y_{obj})$  is the centroid of the detected object. If



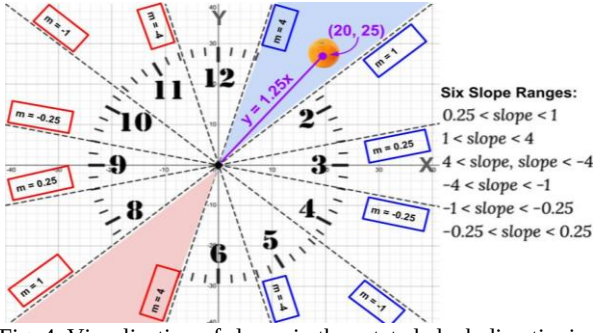


Fig. 4. Visualization of slopes in the rotated clock directioning algorithm: The orange area (top right) marks an object in the 1 o'clock range. All six slope ranges are listed on the right.

$d_{hand,obj}$  is below a predefined threshold, the object is considered within reach, triggering hand-relative guidance.

#### F. System Components and Materials

The framework (Figure 1) was tested on both a computer and a smartphone, ensuring portability for visually impaired users. The smartphone acted as a camera, while the computer handled processing, providing a balance between convenience and computational efficiency. For a fully portable solution, a Raspberry Pi 4 [12] was used, connected to a camera [13] and microphone [14], and powered by a battery pack [15]. This self-contained system enables real-time object detection and interaction, enhancing user independence. The Raspberry Pi was chosen due to its low power consumption and ability to run computer vision models efficiently, making it a cost-effective and practical solution for wearable assistive technology.

### III. SYSTEM IMPLEMENTATION

Sight to Sound Framework pipeline illustrated in figure. 2. An audio input will trigger the framework with a command word to initiate the speech recognition module, allowing it to listen for the object to be searched. Following this, the object detection component will activate the camera to begin searching for the specified object. Once detected, the Rotated Clock Directioning System will provide feedback on the object's location in terms of clock direction, indicating its position relative to the camera, along with the distance of the object from the camera. Detailed implementation information is presented in the following sections.

#### A. System Pipeline

The Sight to Sound Framework is designed to process audio and visual inputs, integrating speech recognition and object detection to provide comprehensive assistance. The process begins with an audio input, which is analyzed using a deep neural network (DNN)-based acoustic model, specifically the VOSK-API (in Section II.B). This model extracts Mel-Frequency Cepstral Coefficients (MFCCs) and utilizes a language model to generate phonetic probabilities mapped to a corresponding lexicon, allowing the framework to accurately interpret spoken commands. Simultaneously, visual input undergoes feature extraction through a modified YOLOv8 convolutional neural network (CNN) model, which identifies objects within the frame using anchor boxes. This step essentially means the framework takes the recognized object name and prepares it for further action, enabling the user to receive specific feedback about what was detected. Depending on the size of the detected object, the framework

assigns either a "land" method or a "grab" method for interaction. If the grab method is selected, a landmark detection model, such as *MediaPipe*, is employed to facilitate the novel hand-relative directioning. This allows the framework to guide users on how to position their hands effectively for grasping the object. Finally, the framework estimates both the distance and the clock direction of the object using relative positioning and novel rotated clock algorithms. The results are then converted into audio output using the gTTS (Google Text-to-Speech) HTML method, providing users with synthesized spoken feedback about the object's location and direction.

#### B. YoloV8 Training

The YOLOv8 model was fine-tuned for enhanced detection of household objects and doors, addressing the limitations of the default COCO-trained version. To enable the 'exit' function, allowing users to ask, "Help me exit" or "Where's the door?", the model was retrained using 4,770 images from Nathaly Espinoza's Roboflow dataset [16]. This optimization improved bounding box precision (mAP), ensuring accurate detection of doors and handles at various distances. The fine-tuned model achieved high accuracy and speed, making real-time objects and door localization reliable and effective for visually impaired users.

#### C. System Implementation on hardware

The framework computer-based system was implemented using a MacBook as the processing unit for algorithms, while a smartphone was held by the visually impaired individual (VI). The smartphone connected to the MacBook via Bluetooth to transmit audio and visual signals captured by its microphone and camera, and to receive directions from the MacBook.

The Sight to Sound framework Raspberry Pi 4-based, utilized the Raspberry Pi 4 as the processing unit to ensure portability. The framework integrates cameras for real-time object detection, battery headspace wires for efficient power management, and a microphone to facilitate user interaction. The camera setup included selecting appropriate models, such as the Raspberry Pi Camera Module, which were connected to the Raspberry Pi 4 to capture video feeds for analysis. The battery headspace wires provided a stable and portable power supply, allowing users to effectively navigate their environment. Additionally, the microphone enabled voice commands and feedback, enhancing user engagement with the framework (see Section II.E).

The software development utilized a lightweight operating system like Raspberry Pi OS, with programming conducted in Python to incorporate libraries for audio processing and computer vision. The interface was designed to provide intuitive auditory feedback, guiding users in locating objects based on real-time video analysis. Through a series of tests,

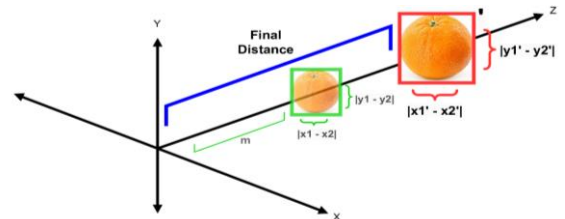


Fig. 5: 3D visualization applies the algorithm to two oranges: a smaller, faded prerecorded one and a larger, live-received one, demonstrating relative distance estimation.

the system's components were verified for functionality and responsiveness, ensuring a reliable user experience. Overall, this integration of hardware and software within the "Object Localization Sight to Sound framework" aimed to empower visually impaired individuals by enhancing their ability to perceive and interact with their surroundings.

#### IV. RESULTS AND DISCUSSION

##### A. Results

To evaluate the framework, we conducted a comprehensive test to assess both the system's and the user's perspectives. From the system's perspective, it detects objects, their direction and the distance from the object to the hand and from the camera to the hand. As illustrated in Figure 7, the system successfully detected both smaller objects (Figure 7(a) & (g)) and larger objects (Figure 7(b)), as well as smaller objects relative to the hand (Figure 7(c), (d), (e)), & (f)). In all of the results, the system was able to detect all objects using the Rotated Clock and Hand Relative Directioning Systems.

From the user's perspective, the evaluation was conducted with four sighted participants who were blindfolded to simulate the experience of visually impaired (VI) users. This approach allowed us to assess the system's real-world performance in assisting with object localization and retrieval. Testing occurred in two phases: first, using an

iPhone camera and laptop in a controlled setting, and second, with a portable Raspberry Pi version under the same conditions. This comparison enabled us to evaluate usability across different platforms.

The assessment focused on two key metrics: search accuracy (object localization) and search speed (retrieval time). Each participant completed 10 tasks—five using the "grab" method for small objects and five using the "land" method for larger ones. Additionally, they used the environment feature three times to detect multiple objects (Table I). Real-life scenarios were simulated, such as searching for an apple and a knife (meal preparation) or a backpack and a car (travel), to evaluate both the system's functionality and the user's interaction.

##### B. Discussion

The framework showcased its efficacy during trials from both the system and the user perspective, simulating the experience of visually impaired (VI) users. A standout feature of this system was the Rotated Clock Directioning System, which proved remarkably effective in providing navigational directions within a three-dimensional (3D) space. This is a significant advancement over traditional assistive technologies that typically operate in a two-dimensional (2D) plane [5],[6]. The clock-based system allows users to visualize their navigational path in an intuitive manner by assigning specific directions—such as "12 o'clock" for straight ahead

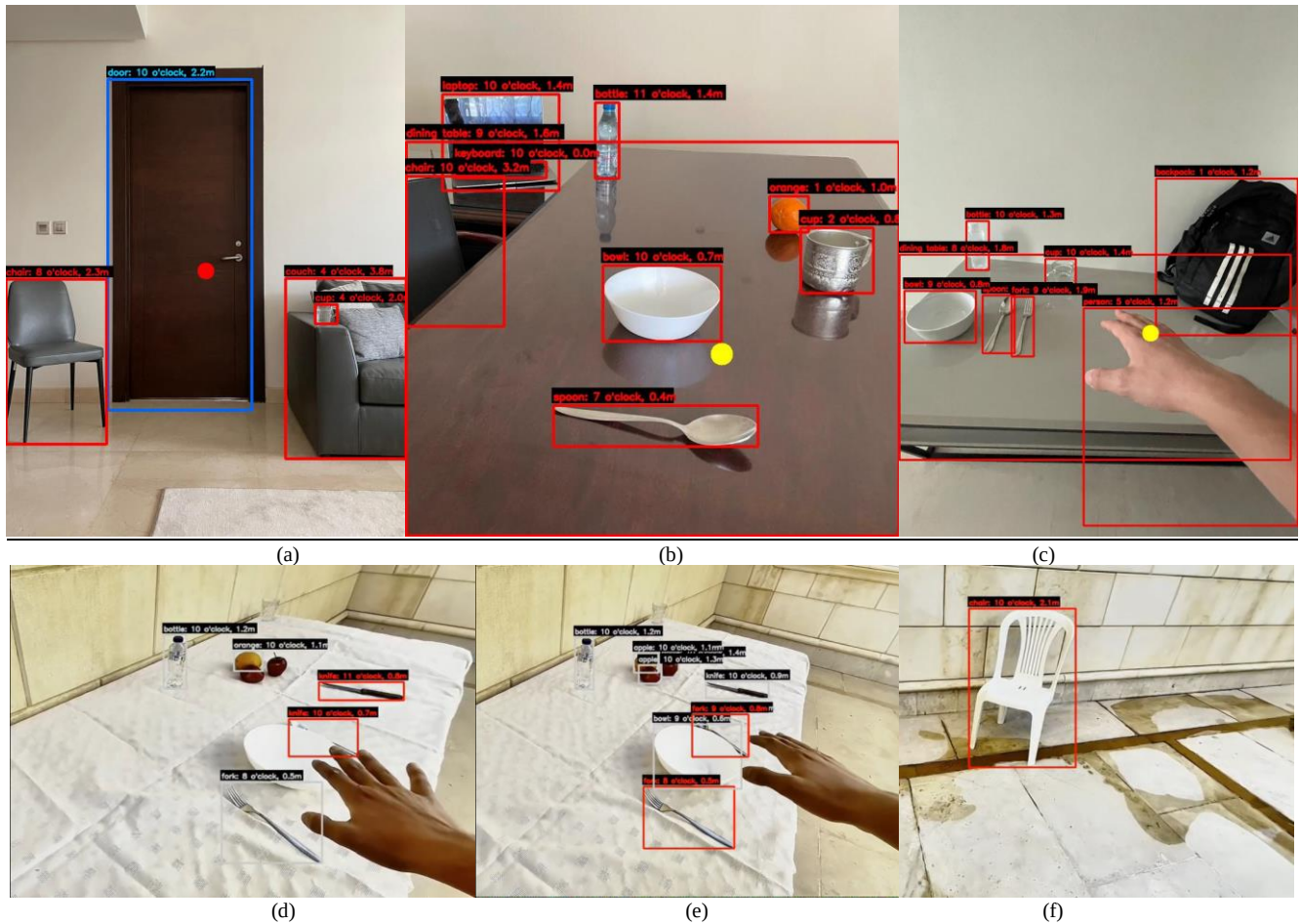


Fig. 6. System's detection perspective: a: smaller objects distance and clock detection based on the frame center, b: larger objects distance and clock detection based the frame center, c: smaller objects distance and clock detection based on the hand center. d: knife detection and tracking as bounding box red, e: fork detection and tracking, f: chair detection and tracking. The framework testing can be found in the following link:

<https://www.youtube.com/watch?v=zGCZzDfeTCs>

TABLE I  
SYSTEM TESTING ON A NUMBER OF SUBJECTS

| User           | G                               | L                               | E                   |
|----------------|---------------------------------|---------------------------------|---------------------|
| 1              | 25s avg.<br>4/5 (80%)           | 25s avg.<br>5/5 (100%)          | 3/3 (100%)          |
| 2              | 26s avg.<br>4/5 (80%)           | 22s avg.<br>5/5 (100%)          | 3/3 (100%)          |
| 3              | 19s avg.<br>5/5 (100%)          | 29s avg.<br>5/5 (100%)          | 3/3 (100%)          |
| 4              | 14s avg.<br>5/5 (100%)          | 20s avg.<br>4/5 (80%)           | 3/3 (100%)          |
| <b>Average</b> | <b>21s avg.<br/>18/20 (90%)</b> | <b>24s avg.<br/>19/20 (95%)</b> | <b>12/12 (100%)</b> |

User: Participants, G: Grab method; L: Land method, F: Environment feature.

enabling them to conceptualize movements more effectively than with vague directional terms like “Upper Right” or “Lower Left.”

The success of the framework was evident in how participants navigated their environment, enhancing their ability to locate and retrieve objects. The immersive experience fostered a sense of confidence and independence, demonstrating the potential for this technology to meaningfully assist VI users in their daily lives. Nevertheless, certain challenges were identified during the testing phase. One notable issue was that users tended to underestimate the extent of their movements in response to directional commands. For instance, User 1 moved only approximately 0.6 [10], [11] meters when instructed to move 1 meter. This indicates a disconnect in interpreting spatial instructions, which resulted in lost session time and reduced user confidence. This behavior is not unexpected, given that this was the participants' first encounter with the device. A viable solution would be to implement a training session to familiarize users with the framework before actual testing. Additionally, enhancing the feedback format to deliver more intuitive output—such as specifying movements in terms of steps rather than meters—could significantly improve user responsiveness. For example, rephrasing commands to “12 o'clock, 1.5 steps” could provide clearer guidance.

The device's camera functionality needs improvement, as participants found its excessive zoom forced awkward hand movements to frame objects, wasting time and reducing the Hand Relative Directioning System's effectiveness, especially for small items. Users extended their hands rigidly, impacting comfort. Simple fixes like adjusting zoom and using relatable distance terms can resolve this. The Raspberry Pi-based system, while slower than a computer due to limited processing power, still detects objects effectively. Future upgrades could use a stronger processor or offload computations to a cloud server via Wi-Fi, creating an IoT framework. Transitioning the Sight to Sound framework to a Swift-based iOS app will enhance mobility, removing laptop dependency. Expanding the object database, improving the interface, and integrating wearables like smart glasses will boost usability, while GPU upgrades will speed up recognition, aiding visually impaired users' independence.

## V. CONCLUSION

This paper presents the Sight to Sound framework for the visually impaired, designed to locate specific objects in various environments, thereby enhancing users' independence and quality of life. The framework integrates

speech recognition and object detection for intuitive navigation, using an innovative Rotated Clock Directioning System and Inferring Distance by Relativity to help users conceptualize movements. Testing has shown its effectiveness while highlighting areas for improvement. By incorporating advanced processing and wearable technologies, we aim to enhance its adaptability. Our work emphasizes the importance of targeted solutions that empower visually impaired individuals to navigate more effectively, contributing to assistive technology innovation.

## REFERENCES

- [1] R. R. A. Bourne et al., "Magnitude, temporal trends and projections of the global prevalence of blindness and distance and near vision impairment: a systematic review and meta-analysis," *Lancet*, vol. 5, no. 9, pp. e888–e879, 2017. DOI: 10.1016/S2214-109X(17)30293-0
- [2] World Health Organization, "Blindness and visual impairment," World Health Organization, [Online]. Available: <https://www.who.int/en/news-room/fact-sheets/detail/blindness-and-visual-impairment>. [Accessed: Mar. 20, 2024].
- [3] A. H. Marion and A. J. Michael, "Assistive Technology for Visually Impaired and Blind People," London, U.K.: Springer, 2008. DOI: 10.1007/978-1-84628-867-8.
- [4] W. M. Elmannai and K. M. Elleithy, "A Highly Accurate and Reliable Data Fusion Framework for Guiding the Visually Impaired," *IEEE Access*, vol. 6, pp. 33029–33054, 2018. DOI: 10.1109/ACCESS.2018.2817164.
- [5] J. F. Oliveira, "The path force feedback belt," *2013 8th International Conference on Information Technology in Asia (CITA)*, 2013. DOI: 10.1109/CITA.2013.6637564.
- [6] L. Frizziero, A. Liverani, G. Donnici, P. Papaleo, and C. Leon-Cardenas, "Smart Cane Developed with DFSS, QFD, and SDE for the Visually Impaired," *Sensors*, vol. 21, no. 3, p. 58, 2021. DOI: 10.3390/inventions6030058.
- [7] Dillon Reis, Jordan Kupec, Jacqueline Hong and Ahmad Daoudi "Real-Time Flying Object Detection with YOLOv8," *arXiv*, 2024. arXiv:2305.09972.
- [8] H. Yu, X. Li, Y. Feng, and S. Han, "Multiple attentional path aggregation network for marine object detection," *Applied Intelligence*, vol. 53, no., pp. 2434–2451, 2023. DOI: 10.1007/s10489-022-03622-0.
- [9] Nickolay V. Shmyrev and other contributors. Vosk Speech Recognition Toolkit: Offline speech recognition API for Android, iOS, Raspberry Pi and servers with Python, Java, C# and Node. <https://github.com/alphacep/vosk-api>, 2020.
- [10] M. P. Murray, R. C. Kory, and S. B. Sepic, "Walking patterns of normal women," *Archives of Physical Medicine and Rehabilitation*, vol. 51, no. 11, pp. 637–650, Nov. 1970, Available: <https://pubmed.ncbi.nlm.nih.gov/5501933/>
- [11] M. P. Murray, A. B. Drought, and R. C. Kory, "WALKING PATTERNS OF NORMAL MEN," *The Journal of Bone and Joint Surgery*, vol. 46, pp. 335–360, Mar. 1964, Available: <https://pubmed.ncbi.nlm.nih.gov/14129683/>.
- [12] 'Raspberry Pi 4' <https://www.raspberrypi.com/products/raspberry-pi-4-model-b/> [Accessed: MAR. 20, 2024].
- [13] 'USB Webcam, 2MP 1080P FHD Center Cam Middle Screen Webcam, AF, FF, MF Meeting Camera with Built In Microphone, 0.14in Lens, for Home Video Conferences' [https://www.amazon.sa/-/en/dp/B0CZQFP956?ref=ppx\\_yo2ov\\_dt\\_b\\_fed\\_asin\\_title](https://www.amazon.sa/-/en/dp/B0CZQFP956?ref=ppx_yo2ov_dt_b_fed_asin_title). [Accessed: Mar. 20, 2024].
- [14] 'GZLMMY Bone Conduction Headphones 3.5mm Wired Headphones with Microphone Suitable for Mobile Phones, Computers, and Tablets, X19B' <https://shorturl.at/LAMyQ>. [Accessed: Mar. 20, 2024].
- [15] 'TalentCell Rechargeable 12V 6000mAh/5V 12000mAh DC Output Lithium Ion Battery Pack for LED Strip and CCTV Camera, Portable Li-ion Battery Bank with Charger, Black (Multi-led Indicator)' <https://shorturl.at/Njqd1>. [Accessed: Mar. 20, 2024].
- [16] "Nathaly Espinoza's, Roboflow" Door detection Object Detection Dataset and Pre-Trained Model by Nathaly Espinoza (roboflow.com) [Accessed: Mar. 20, 2024].